The Shadow Knows: Interaction Records That Outlive Their Carriers in Reversible Second-Order Cellular Automata

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Abstract

In a reversible dynamics no information is destroyed, so the record of an interaction must be somewhere afterwards. We ask where. Using the 256 second-order (Fredkin) extensions of the elementary cellular automata, $x_{t+1} = f(x_t) \oplus x_{t-1}$, we first identify bounded, autonomous, slower-than-light coherences by frozen, outcome-blind searches, and then collide them. For each collision we compute an exact ledger over 64 impact offsets: how much of the information that distinguishes the inputs leaves with the outgoing coherences, and how much stays behind in the background. We then intervene: every resolved coherence is erased, and a fresh probe coherence is launched through the remaining background. Across 91 collision pairs in 25 rules, all 45 pairs whose background returns exactly to vacuum give zero information to the probe, while in all 45 pairs that leave a residue (14 rules) the probe's effect on the future recovers *all* of the information held only by the background (1.00–4.63 bits). In 27 of those pairs the outgoing probe coherence itself carries part of the record away, and in 19 it carries all of it. Every rule that leaves such a "shadow" has an oscillating vacuum. All tests were specified and frozen before their outcomes were computed. Invalid runs and falsified hypotheses are reported alongside the successes.

1 Introduction

Gliders in cellular automata are the standard example of coherent structures that carry and process information: they transmit signals, and their collisions compute [1–4]. Local information dynamics has made this quantitative. Gliders dominate information transfer, collisions dominate information modification, and blinkers and background domains dominate information storage [5–7]. These results are statistical: they average over ensembles of trajectories, and statistical transfer is not the same as causal influence [8, 9].

Here we ask an interventional version of the storage question in a setting where it has an exact answer. In a *reversible* automaton the global map is a bijection, so the information distinguishing two interacting coherences before they meet cannot be lost. After the interaction it must be carried by the outgoing coherences or by the background. If we erase the outgoing coherences entirely, is a record of the interaction still present, and can a later coherence read it? We call such a residual record a *shadow*.

The question is motivated by the Information-Constraint Principle (ICP) [10], which treats physically effective information as interaction-formed constraint on reachable futures that can persist beyond its original carrier. The present paper does not depend on that framework. It reports frozen computational experiments on a complete, well-known rule family, including the rule 37R singled out by Wolfram for its localized structures [11].

Our contributions are:

1. an outcome-blind identification of bounded, autonomous, slower-than-light coherences across all 256 second-order rules;
2. an exact information ledger for collisions, separating what the outgoing coherences carry from what the background holds;
3. a carrier-erasure intervention showing that the background's share is always present and always fully recoverable by a passing probe, with an exact null in every rule whose background returns to vacuum;
4. evidence that the probe coherence itself often carries the recovered record away.

2 Substrate and protocol

Rules. Cells $x_t(i) \in \{0,1\}$ on a ring evolve by $x_{t+1}(i) = f\big(x_t(i-1), x_t(i), x_t(i+1)\big) \oplus x_{t-1}(i)$, where f is any of the 256 elementary truth tables (Wolfram code). Every such rule is exactly invertible, $x_{t-1} = f(x_t) \oplus x_{t+1}$ [11, 12]. The *vacuum* is the orbit of the all-zero state. It is silent when $f(000) = 0$ and a homogeneous oscillation when $f(000) = 1$. All observables are differences from the matched vacuum evolved under the same rule.

Objects. A coherence is a cluster of difference sites, on the union of both time layers and with internal gaps of fewer than three sites, of width at most 8, that is exactly periodic up to translation (pattern and vacuum phase both recur). Its velocity is the exact ratio of shift to period. Its species is the set of states on its orbit.

Discipline. Each test was written as a contract with fixed hypotheses and decision rules. It was committed and tagged in git before the candidate family was evaluated, and the result was then committed as a separate child commit. Exposure to earlier outcomes is disclosed in each contract. Post-hoc analyses are labelled as such. Two runs crashed during serialization before any output; each was refrozen with a disclosed one-line fix. One design was invalidated by its own control and replaced (Section 6).

3 Coherences and their interactions

Formation and autonomy. A detector requiring an isolated core of width at most 8 that persists for 64 ticks (confirmed on a larger ring) found 22 rules in which a single event produces such a core, and 57 rules (243 rule–seed pairs) with seeds of up to four bits. Most of these move at the causal limit. To separate genuine objects from features of larger structures we cut each mature core out, including both time layers, and re-ran it alone in the matched vacuum. 36 of 46 slower-than-light cores (13 rules) survived unchanged, compared with 48 of 194 light-speed cores (Fig. 1). Typical speeds are 1/2 and 1/3. A frozen hypothesis that slow objects require an active vacuum was *falsified* by an exhaustive two-layer seed search: 21 silent-vacuum rules host slow bounded objects, compared with 37 active-vacuum rules.

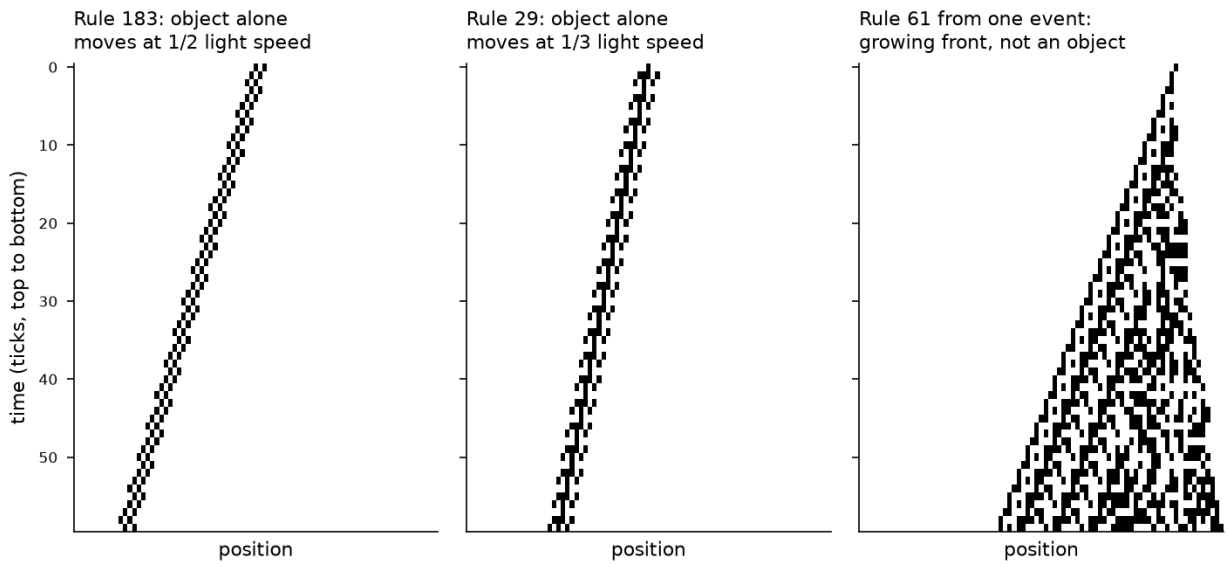


Figure 1: Autonomous coherences cut out of their surroundings and run alone. Left and centre: slower-than-light objects in rules 183 and 29. Right: a single event in rule 61 produces a growing front, not an object. Time runs downward.

Collisions. Head-on collisions of the lowest-seed right- and left-moving objects (200 collisions in 25 rules, eight impact phases each, all verified to make contact) gave 101 elastic outcomes, 20 in which two different species leave at the incoming speeds, 26 exact superpositions, 25 in which the collision opens an expanding domain of the opposite vacuum phase, and 28 unresolved outcomes. None annihilated, as reversibility requires (Fig. 2). In this design every pair had opposite speeds of equal magnitude, so momentum-like conservation could not be tested. Bubbles occurred only in active-vacuum rules, as predicted before the run.

4 The shadow

Ledger. For a fixed rule and pair, the 64 impact offsets form a uniform input set ($H(I) = 6$ bits), and the dynamics is deterministic, so all entropies below are exact for that ensemble. Let O be the description (species, internal state, position) of all outgoing coherences at $t = 600$. Since the full state determines the input, $H(I \mid O)$ is exactly the information the background holds and the coherences do not: the *shadow-only* information.

Intervention. At $t = 600$ every resolved coherence is erased (both layers set to vacuum), leaving only the residue. A probe, an exact copy of the right-moving object's free-flight state, is written at one fixed site outside every shadow for all 64 offsets, so its starting position carries no information about the input. The worlds with and without the probe are run for 1500 further ticks. The *footprint* Y is their difference. The *probe state* Y_p is the set of resolved coherences in the region where they differ. We report $I(I;Y \mid O)/H(I \mid O)$ and $I(I;Y_p \mid O)/H(I \mid O)$.

Results. The null holds exactly: in all 45 pairs whose background returns to vacuum after erasure, both readouts are zero. In all 45 pairs that leave a residue (14 rules) the footprint ratio is exactly 1.

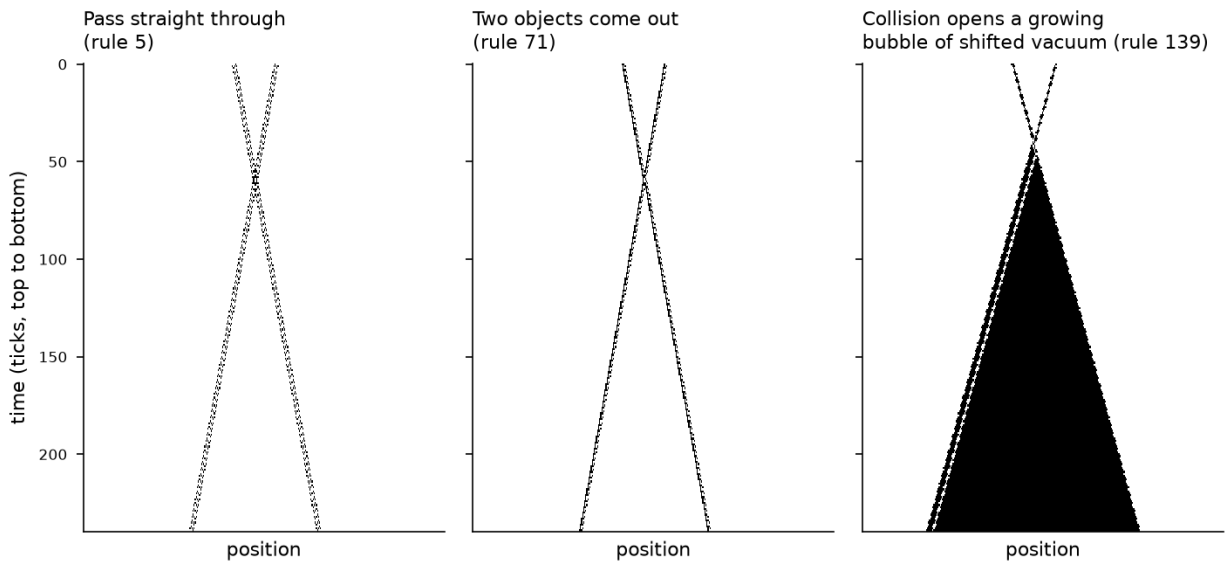


Figure 2: Three collision outcomes: exact pass-through, two coherences out, and a collision that opens a growing domain of the opposite vacuum phase (black).

The probe's passage determines every bit of the shadow-only information, which ranges from 1.00 to 4.63 bits (Fig. 3, left). The record is also often carried out by a coherence: the probe-state ratio is positive in 27 of 45 pairs and equal to 1 in 19, with median 0.60 (Fig. 3, right). A coherence of the probe's own species leaves in a median of 32 of 64 runs. All shadow rules have an active vacuum. An earlier one-pair-per-rule version of the test (12 shadow rules, 13 clean) gave the same footprint result.

5 Related work

Reversible, information-conserving automata and the second-order construction are classical [12–14]. Rule 37R and its structures were described by Wolfram [11], and exactly solvable reversible automata with interacting quasiparticles, notably the reversible Rule 54, are now well understood [15–18]. That work analyses scattering and thermalization, not records left behind once the carriers are gone. Particles as information carriers and domain walls are formalized in computational mechanics [3, 19–22]. The closest precedent is local information dynamics, which identifies background domains as the dominant information storage in irreversible elementary automata [7]. Our contribution is interventional and exact rather than statistical, in line with causal information-flow measures [8, 23]. Collision-based computing uses designed collisions that write stationary memory read by later gliders [4, 24–26]. The shadows here are emergent vacuum-phase residues, not designed localizations. Information left in an environment and read later has also been studied as stigmergy [27] and as outsourced memory through niche construction [28].

6 Limitations and failed runs

The ensembles are finite (64 offsets per pair, at most four pairs per rule) and one-dimensional. The entropies are exact for these ensembles, not capacities over all collisions. The footprint is a

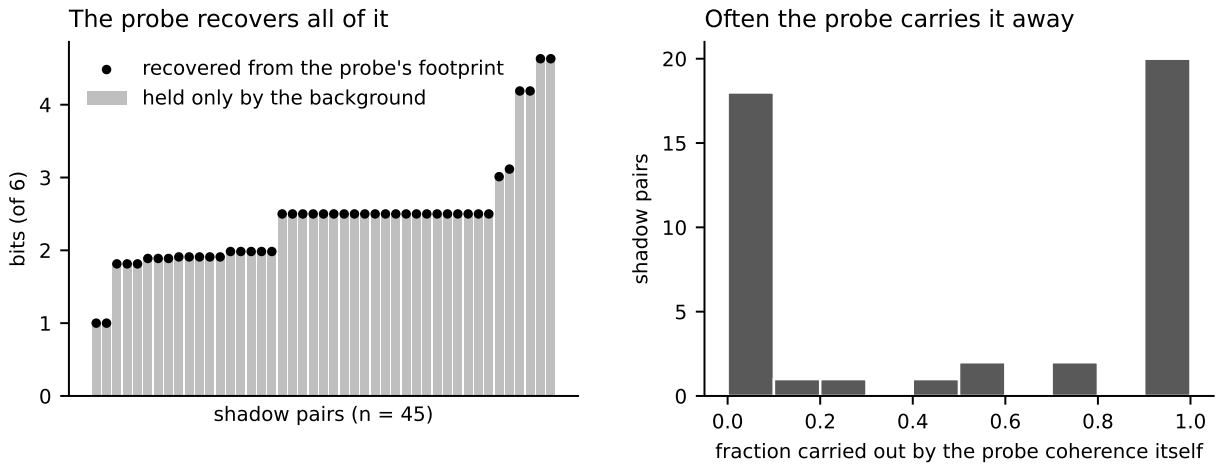


Figure 3: Left: information held only by the background after a collision (bars) and the amount recovered from a probe's footprint after the carriers were erased (dots), for every shadow pair. Right: fraction of that information carried out by the probe coherence itself.

whole-world readout. The probe-state readout is stricter but depends on our object definition (width at most 8, exact periodicity). Every shadow observed so far occurs in an active-vacuum rule, so whether silent vacua can hold shadows remains open.

The path included failures that shaped the final design. An inherited proto-object detector was passed formally by 16 rules that formed only expanding wedges. A collision classifier first counted non-meeting pairs as pass-throughs and mislabelled radiation; a bubble-wall speed measurement was invalid. An object extractor that clustered on one time layer created a spurious null; it was fixed by clustering on both layers. A first shadow design erased carriers before they had separated and failed its own control. The "active vacuum is required for slow objects" hypothesis was falsified. All are recorded in the repository with their commits.

7 Discussion

Reversibility forces the question "where did the information go?" to have an exact answer, and the answer has two parts. Outgoing coherences carry the record mainly relationally, through where they are. When the background is itself dynamic, part of the record is deposited there. That deposit is not inert. With the original carriers erased it fully determines how a later coherence's passage reshapes the future, and often the later coherence leaves carrying it. In the vocabulary that motivated this study, the residue is interaction-formed constraint on reachable futures that has outlived its carriers [10]. Open questions include whether silent vacua can hold shadows, how shadow capacity scales with collision diversity, and whether the result extends to two-dimensional and quantum reversible dynamics.

Reproducibility

All code, frozen contracts, seals and results are in the Iota repository under `src/minimal_realization_reversible`, `docs/ICP_MIR_REVERSIBLE_*`, and `output/minimal_realization/MIR-REVERSIBLE-`

Each contract is tagged `mir-reversible-*-frozen`. Numbers in this paper are generated from the committed result files by `scripts/build_shadow_paper_assets.py`.

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